

Regarding Exam 1 topics: please study the following:

- 1) KVL, KCL, Source transformation, series/parallel, and node voltage methods
- 2) Thevenin/Norton equivalent circuits
- 3) Op amp circuits
- 4) Complex numbers, cartesian $\leftrightarrow$ polar coordinate conversion, sinusoid properties

Simplify
 Series - have same current
 flowing through them
 Parallel - same two nodes

2.41. For the circuit in Fig. P2.41, determine R_{eq} at
 (a) Terminals (a, b)
 (b) Terminals (a, c)
 (c) Terminals (a, d)
 (d) Terminals (a, f)

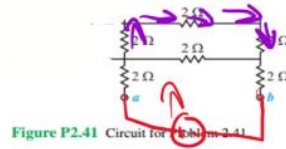
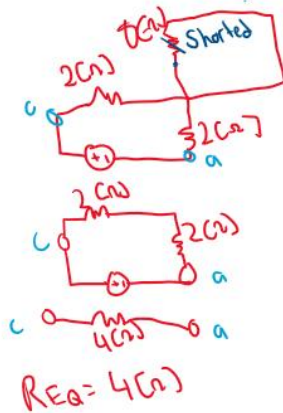
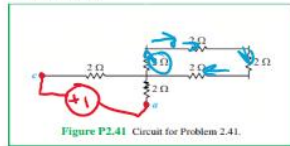
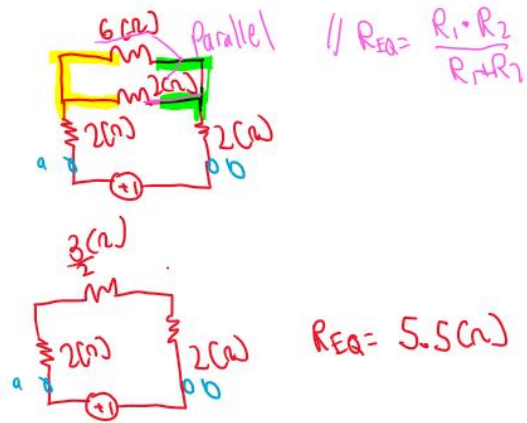


Figure P2.41 Circuit for Problem 2.41



3.71 Find the Thevenin equivalent circuit at terminals (a,b) of the circuit in Fig. P3.71.

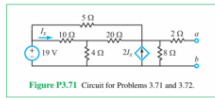


Figure P3.71 Circuit for Problems 3.71 and 3.72.

Thevenin Equivalent



$$V_{OC} = I_{SC} \cdot R_{Th}$$

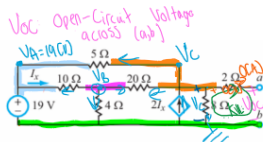
Pick any 2/3.

$$V = IR$$

$$V = 0(2A) + 2(2A) = 0V$$

$$KVL: +V_{OC} - V_C + 0V = 0$$

$$V_{OC} = V_C$$



$$V_A: V_A = 19V$$

$$V_B: \frac{V_B - 19V}{10\Omega} + \frac{V_B - 0V}{4\Omega} + \frac{V_B - V_C}{20\Omega} = 0$$

$$V_C: \frac{V_C - 19V}{5\Omega} + \frac{V_C - V_B}{20\Omega} - 2I_x + \frac{V_C}{2\Omega} = 0$$

Simplify
Collect all like terms

$$\frac{V_C - 19V}{5\Omega} + \frac{V_C - V_B}{20\Omega} - 2I_x + \frac{V_C}{2\Omega} = 0 \times 40\Omega$$

$$LCM: 40\Omega$$

$$9(V_C - 19V) + 2(V_C - V_B) - 80\Omega I_x + 5V_C = 0$$

$$9V_C - 152V + 2V_C - 2V_B - 80\Omega I_x + 5V_C = 0$$

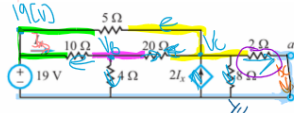
Simplified Equation

$$15V_C - 2V_B - 80\Omega I_x = 152V$$

Norton Equivalent



I_SC: Solve for current through shorted (a,b)



$$I_{SC} = \frac{V_C}{2\Omega}$$

$$V_B: \frac{V_B - 19V}{10\Omega} + \frac{V_B - 0V}{4\Omega} + \frac{V_B - V_C}{20\Omega} = 0$$

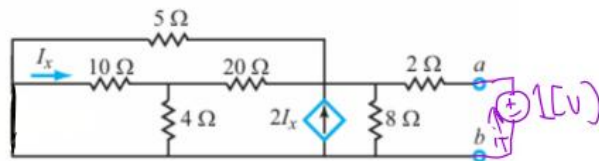
$$V_C: \frac{V_C - 19V}{5\Omega} + \frac{V_C - V_B}{20\Omega} - 2I_x + \frac{V_C}{2\Omega} = 0$$

$$I_x = - \left(\frac{V_B - 19V}{10\Omega} \right) = \frac{19V - V_B}{10\Omega}$$

4 equations, 4 unknowns

$$I_{SC} = 3.96A$$

Resistor Simplification **Only works if no dependent sources*
 Solve R_{EQ} / Test Source Method
 Test Source Method



a)

Solve for it using any method you like.

$$R_{EQ} = \frac{1V}{i_T}$$

$i_T = 0$

If:

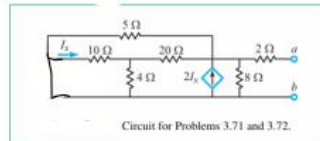
$$V_{OC} = 0V$$

$$I_{SC} = 0A$$

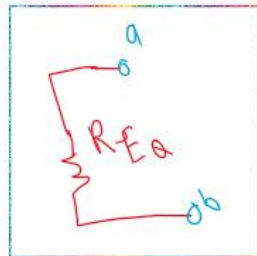
$$V_{OC} = I_{SC} \cdot R_{EQ}$$

$$R_{EQ} = \frac{0}{0} = \text{Undefined}$$

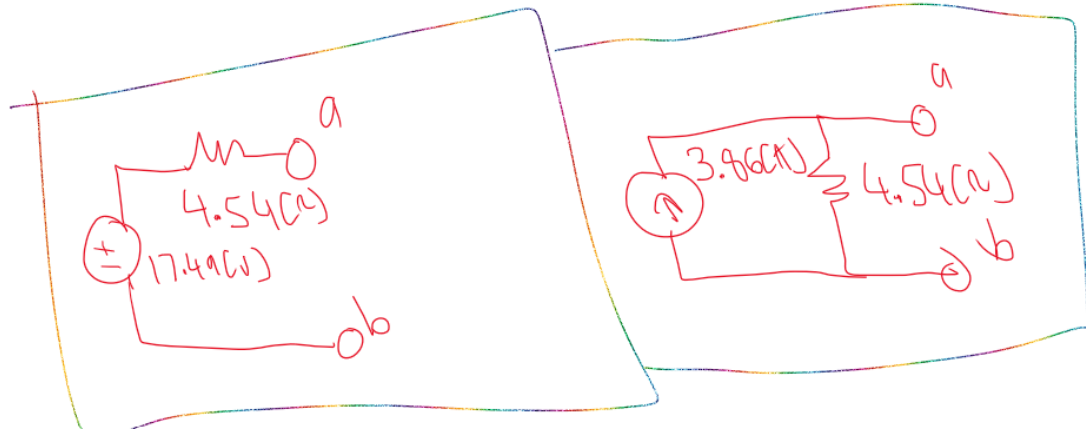
Find the Thévenin equivalent circuit at terminals (a,b) of



Must use R_{EQ} methods directly



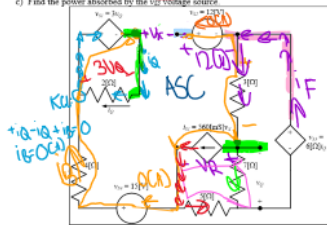
$$R_{EQ} = \frac{V_{oc}}{I_{SC}} = 4.54 \Omega$$



Worry about sign
convention
 $P = VI$ $V = IR$

First Get equations w/ variables of interest V_x, i_F, V_R, i_B, V_Q
Must do KVL for V_x :

3. Using only KVL, KCL, and Ohm's Law (not Node Voltage Method) for the circuit below, solve for the numerical values of the quantities requested.
a) Find i_Q
b) Find the power delivered by the 3V current source.
c) Find the power absorbed by the 2V voltage source.



$$0 + V_x + 12V + i_F \cdot 30\Omega + V_R - 15V + i_B \cdot 4\Omega + 3V_Q = 0$$

$$i_B = 0A$$

Must do KVL for V_R :

$$+V_R + 0.56V_x + 5 + V_Q = 0$$

$$i_Q = -\frac{3V_Q}{2\Omega}$$

$$\text{KCL: } +i_F - 0.56V_x - \left(-\frac{V_Q}{10\Omega}\right) = 0$$

$$+i_F - 0.56V_x + \frac{V_Q}{10\Omega} = 0$$

$$V_x, i_F, V_R, i_B, V_Q$$

a) ✓

Passive Sign Convention =
 $P = VI$

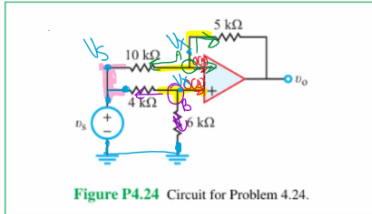
$$b) P_{del. by is_1} = -(V_R \cdot 0.56V_x)$$

Active Sign Convention
 $P_{abs} = -VI$

$$c) P_{abs. by vs_3} = -(6 \cdot i_Q \cdot i_F)$$

Ideal Op-Amps
Virtual Short, No current into Op-Amp input

4.24 For the circuit in Fig. P4.24, obtain an expression for voltage gain $G = v_o/v_s$.



Use KCL, KVL, Ohm's Law.....

to get v_o & v_s in one equation

$$\text{KCL}_A: \left(\frac{v_x - v_s}{10 \text{ k}\Omega} + \frac{v_x - v_o}{5 \text{ k}\Omega} = 0 \right) \quad \text{KCL}_B: \left(\frac{v_x - v_s}{4 \text{ k}\Omega} + \frac{v_x}{6 \text{ k}\Omega} = 0 \right)$$

$$v_x - v_s + 2(v_x - v_o) = 0$$

$$3v_x - v_s - 2v_o = 0$$

$$3\left(v_s \cdot \frac{3}{5}\right) - v_s - 2v_o = 0$$

$$\frac{4}{5}v_s - 2v_o = 0$$

$$\frac{4}{5}v_s = 2v_o$$

$$\frac{4}{10}v_s = v_o$$

VOR:

$$v_x = v_s \cdot \left(\frac{6 \text{ k}\Omega}{4 \text{ k}\Omega + 6 \text{ k}\Omega} \right)$$

$$v_x = v_s \cdot \frac{3}{5}$$

$$\frac{v_o}{v_s} = \frac{4}{10} = \frac{2}{5}$$

Rectangular $\leftrightarrow$ Polar
 $x + jy$ $M \angle \theta$

4. Solve the following equation for m and n . You should assume that m and n are real numbers. The variable n is an angle. There are two sets of solutions, and you should find both sets of solutions. In your solutions, show your units.

$$\frac{(m + 67.3j)}{23 \angle n} = 4.794 - 1.875j \Rightarrow M = \sqrt{4.794^2 + (-1.875)^2} = 5.1476$$

$$\theta = \tan^{-1}\left(\frac{-1.875}{4.794}\right) = -21.36^\circ$$

$A = x + jy = Me^{j\phi}$, where

$$M = \sqrt{x^2 + y^2},$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right),$$

$$x = M \cos \phi, \text{ and}$$

$$y = M \sin \phi.$$

$$(m + 67.3j) \times 23 \angle n = 5.1476 \angle -21.36^\circ \times 23 \angle n$$

$$M_1 \angle \theta_1 \times M_2 \angle \theta_2 = M_3 \angle \theta_3$$

$$M_3 = M_1 M_2$$

$$\theta_3 = \theta_1 + \theta_2$$

Taking real part

$$m = 119.395 \cos(n - 21.36^\circ)$$

$$m = 97.414$$

Taking imaginary part

$$67.3 = 119.395 \sin(n - 21.36^\circ)$$

$$\sin^{-1}\left(\frac{67.3}{119.395}\right) = \sin^{-1}(\sin(n - 21.36^\circ))$$

$$34.6412^\circ = n - 21.36^\circ$$

$$n = 56.0012^\circ$$

$A = x + jy = Me^{j\phi}$, where

$$M = \sqrt{x^2 + y^2},$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right),$$

$$x = M \cos \phi, \text{ and}$$

$$y = M \sin \phi.$$

$$M_1 \angle \theta_1 \times M_2 \angle \theta_2 = M_3 \angle \theta_3$$

$$M_3 = M_1 M_2$$

$$\theta_3 = \theta_1 + \theta_2$$

Phasors and Complex Numbers

Remember $\sin \leftrightarrow \cos$, must be in $\cos()$

Phasors and Complex Numbers

Polar $\rightarrow$ M/L θ , must be in $\cos()$

$$\theta_3 = \theta_1 + \theta_2$$

1. Convert the following time domain signals into phasor form:

- (a) $x(t) = 10 \cos(100t - 30^\circ) = 10 \angle -30^\circ$
- (b) $x(t) = 5 \sin(200t - 45^\circ) = 5 \angle -75^\circ$
- (c) $x(t) = 12 \cos(\omega t - 90^\circ) = 12 \angle -90^\circ$

2. Convert the following phasors back to time domain expressions:

- (a) $\hat{V} = 8 \angle 45^\circ = 8 \cos(\omega t + 45^\circ)$
- (b) $\hat{I} = 6 \angle -120^\circ$

3. Perform the following additions:

- (a) $4 \angle 30^\circ + 3 \angle 40^\circ = 6 \angle -15^\circ$
- (b) $10 \angle 20^\circ + 5 \angle 90^\circ = 10 \cos(20^\circ) + j 5 \sin(90^\circ) = 9.397 + j 5$

4. Perform the following subtractions:

- (a) $8 \angle -25^\circ - (1 + j4)$
- (b) $8 \angle 45^\circ - 4 \angle -45^\circ$

5. Convert these phasors from polar to rectangular form:

- (a) $10 \angle 30^\circ$
- (b) $5 \angle -90^\circ$
- (c) $12 \angle 150^\circ$

6. Convert these phasors from rectangular to polar form:

- (a) $3 + j4$
- (b) $-3 + j5$
- (c) $-2 - j0$

M/L

$$M = \sqrt{3^2 + 4^2} = 5$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) = 56.3^\circ$$

$$\theta = \tan^{-1}\left(\frac{-6}{-2}\right) = 180^\circ$$

$$\theta = \tan^{-1}\left(\frac{-6}{-2}\right) = 180^\circ$$

$$M \angle \theta = 5 \angle 135^\circ$$

7. Multiply the following phasors:

- (a) $4 \angle 30^\circ \cdot 3 \angle 40^\circ = 12 \angle 70^\circ$
- (b) $(2 + j3)(1 - j2) = 2 \angle 45^\circ \cdot \sqrt{5} \angle 71.56^\circ = \sqrt{10} \angle 26.56^\circ$

8. Divide the following phasors:

- (a) $\frac{2 \angle 15^\circ}{1 \angle 45^\circ} = 2 \angle -30^\circ$
- (b) $\frac{10 \angle 90^\circ}{2 \angle 30^\circ} = 5 \angle 60^\circ$

$+$ $- \rightarrow$ Rectangular form

$\times \div \rightarrow$ Polar form

$\times^y \rightarrow$ Polar form

$$M = \sqrt{2^2 + 2^2} = \sqrt{8}$$

$$\theta = \tan^{-1}\left(\frac{2}{2}\right) = \tan^{-1}(1) = 45^\circ$$

$$M = \sqrt{1^2 + (-3)^2} = \sqrt{10} \quad \theta = \tan^{-1}\left(\frac{-3}{1}\right) = -71.56^\circ$$

$$\text{Multiply } (1 + j)(3 - 2j)$$

$$\text{Step 1/2 } y(5 + 2j) + (7 - 9j)$$

$$\text{Multiply } (2 + 3j)(2 - 3j)$$

$$\text{Multiply } (1 + j)(3 - 2j)$$

$$\text{Find the Conjugate } (-4 - 7j)$$

$$\text{Divide by multiplying by the conjugate}$$

$$\text{Convert to polar form } z = 6 + 4j$$

$$z = 2 \angle 15^\circ \quad w = 3 \angle 30^\circ \text{ find } z \cdot w$$

$$\text{Divide by multiplying by the conjugate}$$

$$\text{Convert to polar form } z = 6 + 4j$$

$$z = 2 \angle 60^\circ \quad w = 3 \angle 20^\circ \text{ find } z/w$$

$$A \cos(\omega t + \phi)$$

A circuit has an RMS value of 4 A, a frequency of 60 Hz, and lags the reference by 45°

$$A = A_{\text{RMS}} \sqrt{2} = 4 \sqrt{2}$$

$$\omega = 2\pi f = 60 \cdot 2\pi = 120\pi \text{ rad/s}$$

$$\phi = -45^\circ$$

$$A = 5.657 \cos(120\pi t - 45^\circ)$$

A signal has a peak-to-peak value of 10 V, oscillates at 1 kHz, and has a phase shift of -180°

$$A = \frac{V_{\text{p-p}}}{2} = 5 \text{ V}$$

$$\omega = 1000 \text{ rad/s}$$

$$\phi = -180^\circ$$

$$A = 5 \cos(1000t - 180^\circ)$$

A sinusoidal voltage source has a maximum value of 12 A at $t = 0$ and oscillates at 60 rad/s

$$A = 12 \text{ A}$$

$$\omega = 60 \text{ rad/s}$$

$$\phi = 0^\circ$$

$$A = 12 \cos(60t)$$

$$A = 12 \cos(60t)$$

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